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System and method for integrating data from a remote server with a client application

Abstract

The electronic device may invoke a host application. The device may display a first host user interface on the display, the first host user interface including a first host user interface component associated with a child application. In response, the device may invoke, using the host application, the child application executed at the electronic device. The device may transmit, using the host application, a request for data associated with a child user interface component to the child application. The device may transmit, using the child application, the request for data associated with the child user interface component to a child application server, wherein the data associated with the child user interface component is inaccessible by the host application. The device may receive, using the child application, the data associated with the child user interface component from the child application server. The device may display the child user interface component.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/387,161 filed Jul. 28, 2021, which is a continuation of U.S. patent application Ser. No. 17/100,132, now U.S. Pat. No. 11,107,057, which is a continuation of U.S. patent application Ser. No. 15/297,589 filed Oct. 19, 2016, now U.S. Pat. No. 10,776,770, which claims the benefit of U.S. Provisional Patent Application 62/243,316 filed Oct. 19, 2015, which are incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention relates to computer systems in which and, more particularly, to computer systems employed in connection with integrating data with client applications.

SUMMARY

(2) In one embodiment there is a method implemented at an electronic device configured to display user interfaces and user interface components on a display of the electronic device. The device may invoke a host application executed at the electronic device. The device may display, using the host application, a first host user interface on the display, the first host user interface including a first host user interface component associated with a child application separate and distinct from the host application. The device may, in response to a selection of the first host user interface component in the first host user interface by an end user, (i) invoke, using the host application, the child application executed at the electronic device; (ii) transmit, using the host application, a request for data associated with a child user interface component to the child application; (iii) transmit, using the child application, the request for data associated with the child user interface component to a child application server connected to the child application, wherein the data associated with the child user interface component is inaccessible by the host application; (iv) receive, using the child application, the data associated with the child user interface component from the child application server; and (v) display the child user interface component on the display.

(3) In one embodiment, there is an electronic device comprising a display; one or more memory units each operable to store at least one program; and at least one processor communicatively coupled to the one or more memory units, in which the at least one program, when executed by the

at least one processor, causes the at least one processor to perform one or more steps. The processor may invoke a host application executed at the electronic device. The processor may display, using the host application, a first host user interface on the display, the first host user interface including a first host user interface component associated with a child application separate and distinct from the host application. The processor may, in response to a selection of the first host user interface component in the first host user interface by an end user: (i) invoke, using the host application, the child application executed at the electronic device; (ii) transmit, using the host application, a request for data associated with a child user interface component to the child application; (iii) transmit, using the child application, the request for data associated with the child user interface component to a child application server connected to the child application, wherein the data associated with the child user interface component is inaccessible by the host application; (iv) receive, using the child application, the data associated with the child user interface component from the child application server; and (v) display the child user interface component on the display.

(4) In one embodiment, there is a non-transitory computer readable storage medium having stored thereon computer-executable instructions which, when executed by a processor, perform one or more steps at an electronic device configured to display user interfaces and user interface components on a display of the electronic device. The processor may invoke a host application executed at the electronic device. The processor may display, using the host application, a first host user interface on the display, the first host user interface including a first host user interface component associated with a child application separate and distinct from the host application. The processor may, in response to a selection of the first host user interface component in the first host user interface by an end user: (i) invoke, using the host application, the child application executed at the electronic device; (ii) transmit, using the host application, a request for data associated with a child user interface component to the child application; (iii) transmit, using the child application, the request for data associated with the child user interface component to a child application server connected to the child application, wherein the data associated with the child user interface component is inaccessible by the host application; (iv) receive, using the child application, the data associated with the child user interface component from the child application server; and (v) display the child user interface component on the display.

(5) In some embodiments, the child user interface component is a child user interface displayed and controlled by the child application and data displayed on the child user interface and data received via the child user interface is inaccessible by the host application.

(6) In some embodiments, the host application is a retailer-specific application, the child application is a financial services application, the child user interface is a financial services user interface displayed and controlled by the financial services application, the financial services user interface displays data associated with a retailer-specific financial account associated with the end user, and the data associated with the retailer-specific financial account associated with the end user is inaccessible by the retailer-specific application.

(7) In some embodiments, displaying the child user interface component includes: providing, using the child application, the data associated with the child user interface component to the host application; generating the child user interface component based on the data associated with the child user interface component; and displaying, using the host application, the child user interface component on the display.

(8) In some embodiments, the host application is a retailer-specific application, the child application is a financial services application, and the data associated with a child user interface component includes data associated with a retailer-specific credit account. In response to a determination that the data associated with the retailer-specific credit account meets end-user-specific offer criteria, displaying the child user interface component includes displaying a first offer to the end user to use the retailer-specific credit account to make a purchase; and in response to a determination that the data associated with the retailer-specific credit account does not meet end-

user-specific offer criteria, displaying the child user interface component includes displaying a second offer to the end user to request a credit limit increase for the retailer-specific credit account to make a purchase.

(9) In some embodiments, the child application is configured to receive personal information provided by the end user, where the personal information is inaccessible by the host application, and the request for data associated with the child user interface component transmitted from the child application includes the personal information provided by the end user.

(10) In some embodiments, the data associated with the child user interface component received by the child application from the child application server is based on the personal information provided by the end user that is inaccessible by the host application.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The foregoing summary, as well as the following detailed description of embodiments of the invention, will be better understood when read in conjunction with the appended drawings of an exemplary embodiment. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown.

(2) In the drawings:

(3) FIG. 1 is a block diagram illustrating an exemplary computer architecture used in connection with carrying out the present invention.

(4) FIGS. 2A-2T illustrate exemplary user interfaces for extending functionality of a host application on an electronic device by providing additional financial services using data that is inaccessible to the host application, according to at least some embodiments of the invention.

(5) FIGS. 3A-3C illustrate exemplary user interfaces for extending functionality of a host application on an electronic device by providing user-specific offers on the host application using data that is inaccessible to the host application, according to at least some embodiments of the invention.

(6) FIG. 4 illustrates a flow diagram illustrating a method performed by an electronic device for extending the functionality of a host application based on data that is inaccessible to the host application, in accordance with some embodiments of the invention.

DETAILED DESCRIPTION

(7) As a result of the ubiquity of mobile phones and mobile applications, every day thousands of data exchanges occur online that involve personal data. These data exchanges may occur between a user and a mobile application or between a mobile application and a backend server. These exchanges of personal data can be helpful. For example, a user may purchase goods or services or access banking information using different mobile applications. However, each time a data exchange occurs, the personal data of a user is exposed to third parties resulting in an increased chance of the personal data being compromised.

(8) For example, if a user attempts to purchase a product via a retailer's mobile application, the user will generally submit personal banking information, such as a credit/debit account number to the retailer's mobile application. In turn, the mobile application will transmit the personal banking information to the retailer's transaction processing servers. From there, the retailer's transaction processing servers will verify the personal banking information with the bank issuing the credit/debit card and complete the transaction. During this process, the user's personal banking information is exposed to the retailer's mobile application and the retailer's transaction processing servers. While most retailers act in good faith, each exposure presents a risk that the user's personal information will be compromised.

(9) In a further example, retailers may desire to enhance their mobile applications by customizing

the user experience. For example, the retailer may want to offer certain discounts or coupons based on a user's personal information. Generally, the retailer can utilize user information acquired from the user directly, such as user demographics information or prior shopping behavior tracked and recorded via the mobile application. However, having access to additional user personal information, such as available credit on a financial account that was not acquired directly from the user, can substantially increase the likelihood that the user makes a purchase using the retailer's mobile application by allowing a retailer to provide targeted offers to the user. A retailer could request access to this personal information from the user directly via the retailer's mobile application by requesting personal identification information of the end user to access the financial information. However, again, while most retailers act in good faith, each exposure of personal information presents a risk that the user's personal information will be compromised.

(10) In a further example, retailers may offer retailer-specific credit cards (i.e., store-branded credit cards that can only be used at that store) to its customers. These credit cards are generally managed by third-party financial institutions (e.g., private label credit cards). In conventional systems, a user may need to access both a retailer application to access retailer-specific services and a separate financial institution application to manage the retailer-specific credit account hosted by the financial institution. Navigating between different applications can be a time-consuming process, which could result in fewer users adopting the retailer-specific credit cards.

(11) At least some embodiments of the present invention are disclosed that protect a user's personal information from being compromised by a host application (e.g., a retailer application) by integrating features from a child application (e.g., a financial institution application) into the host application while simultaneously preventing exposure of a user's personal information to the host application. In some embodiments, the host application invokes a child application that accesses and/or processes the personal information of the user. The child application then provides data to the host application for display on the display interface of a client device while also segmenting the user's personal information from the host application. These embodiments describe an integration concept where a child application is executed by a host application and control of a user interface is transferred between the host application and the child application. In these embodiments, the child application is encoded binary, not source code, meaning that the second application cannot be modified by the host application before, during or after execution. These embodiments generally improve the security functionality of a computer by providing a series of controlled access points in a computer application for a user to provide sensitive personal information without exposing the user's personal information to an unknown and unverified third party. One benefit of allowing the child application to display data and interact with the user is for enhanced security. In this modality, no data is transferred between the host and child application. Therefore, the child application may display sensitive content to the user in the form of personal identification information (PII) data or Payment Card Industry (PCI) data, without storing or transmitting such data to the host application. In addition, the child application can implement its encryption and security model for temporarily storing and transmitting the data to third parties. These embodiments also ensure that the host application will not break the security model established by the child application developer. Such an implementation also minimizes work performed by the host application developer because the developer does not need to develop corresponding host application user interfaces for child application data. Overall, these embodiments lead to a lower burden of testing and oversight into the host application development process and resulting source code.

(12) This type of integration can be used in a number of different environments or industries, including retailing, manufacturing, health care, or other environments that can utilize a customer-facing application. For example, using this functionality, a retailer can provide a native credit experience that seamlessly inserts credit functionality into a retailer shopping application without disrupting the branding of the retailer shopping application. Other examples include digital service providers who provide solutions that can be embedded within third party applications, and loyalty

providers who can extend advertising to other retailer's applications. As another example, a car dealer with multiple dealerships might provide a dealership parent application to its customers while also using a car manufacturer child application, where the car manufacturer child application provides the details on cars for display on the user device when the user initiates use of the dealership parent application.

(13) Exemplary System Embodiments

(14) FIG. 1 illustrates an exemplary block diagram of a system **10** for extending functionality of a host application using data that is inaccessible to the host application, according to at least some embodiments of the invention. In this example, the system **10** includes a client device **100**, a host application server **138** and a child application server **140** to interface with an end user **150**.

(15) The client device **100** may be any computing device configured to interface with the end user **150** and/or the servers **138** and **140**. Examples of the client device **100** may include a smart phone, tablet or a personal computer, among others.

(16) In the embodiment as shown, the client device **100** includes a user interface **102** to receive inputs from and display data to the end user **150**. Examples of the user interface **102** include touch screens, and monitors with peripheral components such as a keyboard and mouse, among other things.

(17) In the embodiment as shown, the client device **100** includes a host application **110** to generate graphical user interfaces for display to the end user **150**. As used herein, the host application **110** may be any executable computer application (other than computer applications executed by the operating system of the client device **100**) that interacts with the end user **150** via a user interface. For example, in some embodiments, the host application **110** may be a retailer application to execute functionality offered pursuant to an application of a retailer. By way of example, the retailer application **110** may be programmed to execute functionality involving shopping, commerce, or a store locator, among others.

(18) The host application **110** may receive user inputs from the end user **150** via the user interface **102**. Alternatively, the host application **110** may request data from the host application server **138** and/or a child application **120** for display via the user interface **102**.

(19) The host application server **138** may be any computing device configured to manage access to a centralized resource or service in a network. In some embodiments, the host application server **138** may provide data resources to facilitate the host application **110** with providing graphical user interfaces to the end user **150**. For example, the host application server **138** may host retailer data used for shopping or commerce, among others. The retailer data may include retailer user interfaces displayable on the user interface **102** to aid the end user **150** during a shopping experience with the host application **110**. The retailer data may also include product information, such as product description and pricing information.

(20) In the embodiment as shown, the client device **100** includes the child application **120**, referenced above. As used herein, the child application **120** may be any executable computer application (other than computer applications executed by the operating system of the client device **100**) invoked by the host application **110** that interacts with the end user **150** via a user interface. In some embodiments, the child application **120** provides additional functionality (e.g., providing services) that is not provided by the host application **110**. For example, in some embodiments, the child application **120** may be a financial services application that provides financial services that extend functionality of an application of a retailer. By way of example, the child application **120** may be programmed to execute additional financial services of the host application **110** specific to the end user **150**, such as accessing and displaying financial account information, transaction history, credit or fund availability on a bank account, payment processing, and digital receipt information, among others.

(21) In some embodiments, the child application **120** may request data associated with the personal information of the end user **150** from a child application server **140** to integrate with graphical user

interfaces. The child application server **140** may be any computing device configured to manage access to a centralized resource or service in a network. In some embodiments, the child application server **140** may provide data resources to facilitate the child application **120** with providing graphical user interfaces to the end user **150**. For example, the child application server **140** may be associated with a financial institution and may host, for the end user **150**, personal information including financial account information, transaction history, credit or fund availability on a bank account, payment processing, digital receipt information from a financial institution.

(22) In some embodiments, when requested by the host application **110**, the child application **120** controls the display on the user interface **102**. In these embodiments, the child application **120** may display personal information of the end user **150** inaccessible to and unmodifiable by the host application **110**. For example, the child application **120** may provide graphical user interfaces related to accessing financial account information or purchasing a product offered by the retailer using the financial account information. Examples of graphical user interfaces provided by the child application **120** include shopping cart or payment UIs that are specifically linked to a financial account of the end user **150** with a specific financial institution. With this functionality implemented, the child application **120** can ensure that the personal information of the end user **150** will not be compromised by the host application **110**, either inadvertently or intentionally, thereby improving the security functionality of the computer. Ultimately, the fewer the number of companies that have access to personal information, the less chance that the information can be compromised.

(23) In some embodiments, when requested by the host application **110**, the child application **120** may provide data associated with personal information that is otherwise inaccessible to the host application **110**. In these embodiments, the host application **110** is providing a graphical user interface to the end user **150**. For at least one of the user interface objects displayed to the end user **150** on the graphical user interface, the child application **120** provides data associated with the graphical user interface object. The data may be a graphical user interface object itself or data that can be processed by the host application **110** to generate a graphical user interface object.

(24) In some embodiments, the data provided by the child application **120** to the host application **110** includes data based on, associated with or generated from personal information of the end user **150** that is inaccessible to the host application **110**. For example, the host application **110** may desire to offer a discount or coupon that, if redeemed by the end user **150**, reduces the purchase price for an object. The discount or coupon may be based on personal information, such as credit availability, personal banking information, end user behavioral information (e.g., prior purchasing habits), and/or end user demographic information that is inaccessible to the host application **110** without the child application **120**. For example, if a user has \$100 of available credit on a financial account, a retailer may desire to offer the end user a \$10 off coupon for any purchase between \$50 and \$100 dollars using a host application **110**. This type of offer may be enticing for a user with \$100 of available credit, but in other cases, where the user less than \$50 of available credit, the user will not be able to redeem the offer because the user does not have access to sufficient funds to redeem the offer. In this case, the retailer may desire to offer a different discount, such as \$5 off a purchase of \$30 or more. By having access to available credit information on a financial account provided by the child application **120**, the retailer can provide user-specific offers to the end user **150** that have a greater likelihood of being redeemed. This type of functionality can be implemented by a host application **110** for a retailer without the end user **150** having to provide any personal information to the retailer via the host application **110**, such as personal identification information to access the financial account information. By preventing a retailer's access to personal information, while still allowing the retailer to provide targeted offers to the end user, the end user can protect personal information while still receiving enhanced functionality (e.g., targeted offers) from the retailer. These embodiments represent an improvement to the security functionality of a computer because the personal information of the end user **150** cannot be

compromised by the host application **110**. On the other hand, conventional implementations risk compromising personal information of users by having host applications directly request access to a user's personal information in order to provide targeted offers.

(25) The host application **110** and child application **120** exchange data with the user interface **102** via connection lines **132** and **134**, respectively. Using the connection lines **132** and **134**, the host application **110** and child application **120** can control the user interface **102** by transmitting user interface data to the user interface **102**. Upon receipt, the user interface **102** generates a graphical user interface on a display for the end user **150**. The user interface **102** may receive user inputs from the user and transmit data representative of the user inputs to the host application **110** and the child application **120** via connection lines **132** and **134**, respectively.

(26) The host application **110** and the child application **120** exchange data via connection line **130**. Using the connection line **130**, the host application **110** can request the child application **120** to control of the user interface **102**. As discussed herein, by transferring control of the user interface **102**, the child application **120** can interact with the end user **150** so that the end user **150** can provide personal information without fear of access by the host application **110**. Using the connection line **130**, the host application **110** can request data associated with the personal information of the end user **150** from the child application **120**. The child application **120** can provide data associated with the personal information of the end user **150** to the host application **110**. In some embodiments, connection line **130** provides bidirectional communication between the host application **110** and the child application **120** and resides entirely on the client device **100**.

(27) The host application **110** connects to the host application server **138** and the child application **120** connects to the child application server **140** via network **136**. Network **136** connects the devices or components by carrying signals. Network **136** may be implemented using wire or cable, fiber optics, a phone line, a wireless link, a cellular phone link, a radio frequency link, or any other suitable communication channel. For instance, network **136** may be implemented using a combination of channels. Network **136** may be implemented as an intranet and/or an internet.

(28) Exemplary User Interfaces

(29) FIGS. 2A-2L illustrate exemplary user interfaces for extending functionality of a host application on an electronic device by providing additional services using data that is inaccessible to the host application, according to at least some embodiments of the invention.

(30) In these embodiments, the host application **110** is a retailer application that sells goods or services to customers. In this example, the retailer application **110** is associated with a fictional retailer known as "Rock Red," a clothing company that sells clothing-related merchandise. The retailer application **110**, in conjunction with host application server **138**, provides one or more user interfaces to the end user **150** via user interface **102**.

(31) In these embodiments, the retailer application **110** is implemented on a mobile device (i.e., the client device **100**). To start the process, the end user **150** opens retailer application **110**, for example, by selecting an icon (not shown) representing the retailer application **110** on the end user's mobile device **100**. In response, the retailer application **110** displays a home page, as shown in exemplary FIG. 2A, on a user interface **102**. In this example, the user interface **102** is a touchscreen.

(32) The end user **150** may navigate through one or more user interfaces of retailer application **110** to shop for clothing merchandise that the end user **150** is interested in purchasing. As shown in FIG. 2B, the user may navigate to a user interface of the retailer application **110** showing a shirt that the user would like to purchase. FIG. 2B includes an image **204** of the shirt and a selectable purchase button **206**. After the end user **150** decides to purchase the shirt, the end user **150** selects the purchase button **206**.

(33) In response, the retailer application **110** displays a checkout user interface, as shown in exemplary FIG. 2C, to complete the transaction. In some embodiments, the retailer application **110** may include financial services functionality, such as a checkout page including a first checkout

button **208** that, if selected, allows the end user **150** to provide financial account information via subsequent user interfaces to complete the purchasing transaction. Once the end user **150** provides the financial account information, the retailer application **110** displays a confirm transaction user interface, as shown in exemplary FIG. 2D. Upon selection of the confirm checkout button **211** by the end user **150**, the transaction is verified and completed. However, as described herein, if the end user **150** provides personal information (e.g., a credit card number) to the retailer application **110**, it is possible that the personal information could be compromised. To address this concern, in some embodiments, the retailer application **110** can provide an alternative checkout process that includes extended financial services functionality. In these embodiments, the retailer application **110** provides a second checkout button **210**, as shown in FIG. 2C, that allows a user to complete the purchasing transaction using a retailer-specific credit card linked to a financial institution. In this example, the retailer-specific credit card is a “RockRed Card.” Upon selection of the second checkout button **210**, the retailer application **110** invokes a child application **120** that provides extended financial services of the financial institution to complete the purchasing process. In this embodiment, the child application **120** is an exemplary financial services application hosted by the financial institution. The retailer application **110** then transfers control of the touchscreen **102** to the financial services application **120**.

(34) In response to the transfer of control from the retailer application **110**, the financial services application **120** requests data associated with a financial services user interface from child application server **140**. In this example, the child application server **140** is a financial services server. The financial services server **140** provides data associated with a retailer-specific financial account of the end user **150** hosted by a financial institution. This data includes user interfaces to complete a transaction.

(35) Initially, as shown in FIG. 2E, the financial services application **120** displays a login user interface on the touchscreen **102**. The user is prompted to enter personal identification information, i.e., a username at field **212** and a password at field **214**. Upon completion, the user can select a login button **216** to login to the financial services application **120**. While this embodiment includes a login user interface, it is contemplated that the login user interface may be optional, as a user may have previously provided personal identification information.

(36) Logging in to the financial services application **120** allows for the secure access of the end user's information stored by the financial services server **140** of the financial institution and use of such information in connection with the retailer application **110**. More particularly, financial services application **120** makes a call to the financial services server **140** of the financial institution. The call includes user credentials, i.e., a user name and password inputted during log in, and an identifier of the mobile device **150** being used by the end user. The security comes from both the user name and password and the ability to identify key attributes about the end user **150**.

(37) In some embodiments, the financial services application **120** may use device fingerprinting (e.g., accessing a device identifier) to provide an additional factor for authentication. In some embodiments, the financial services application **120** auto generates the device identifier and stores it locally within an encrypted storage.

(38) In some embodiments, if the end user **150** has already logged into the financial services application **120**, then the financial services application **120** can forego display of the login user interface.

(39) After the end user **150** has logged in and has been authenticated, the financial services application **120** displays a confirm checkout user interface, as shown in FIG. 2F. The confirm checkout user interface includes a “Confirm Checkout using RockRed Card” confirmation button **218** that, when selected, completes the purchase transaction using the financial services application **120**. To complete the purchase transaction, the financial services application **120** transmits personal information of the end user **150**, such as personal identification information (e.g., username and password) or personal account information (e.g., credit card account information) to the financial

services server **140**, where the purchase transaction is verified.

(40) By completing the purchase transaction using the financial services application **120**, the end user **150** avoids providing personal information (e.g., credit card information) to a retailer via the retailer application **110**, thereby improving the security functionality of the computer by protecting the user's personal information from being compromised. These embodiments represent an improvement over conventional applications that merely request the user to provide the personal information (e.g., credit card information) to complete the transaction.

(41) Besides the checkout functionality, different embodiments of the financial services application **120** include additional financial services functionality. The additional financial services functionality can be accessed via a multitude of different methods. In one example, the additional financial services functionality is accessed by the end user by selecting a credit card link **219**, as shown on FIG. 2G.

(42) There are a number of different examples of additional financial services functionality. For instance, in some embodiments, the financial services application **120** displays functionality to view an account balance for an retailer-specific account of the end-user **150** provided by the financial institution (see, for example, FIG. 2H). In a further example, in some embodiments, the financial services application **120** displays functionality to view an transaction history for the retailer-specific account of the end-user **150** provided by the financial institution (see, for example, FIG. 2I). In a further example, in some embodiments, the financial services application **120** displays functionality to pay the account balance for the retailer-specific account of the end-user **150** provided by the financial institution (see, for example, FIG. 2J). In a further example, in some embodiments, the financial services application **120** displays offers associated with the retailer-specific account of the end-user **150** provided by the financial institution (see, for example, FIG. 2K). In a further example, in some embodiments, the financial services application **120** implements functionality to display a digital scan-able credit card associated with the retailer-specific account of the end-user **150** provided by the financial institution (see, for example, FIG. 2L).

(43) FIGS. 2M-2T illustrate exemplary user interfaces for applying for new credit usable for purchases in the host application by using data that is inaccessible to the host application, according to at least some embodiments of the invention. As illustrated in FIG. 2M, after the user navigates through the host application to the child application, the user is presented with a user interface that includes an offer for credit (e.g., a Rock Red Store Card). In some embodiments, the child application may display opt-in options for data sharing (e.g., "Apply with Synchrony Bank" selectable icon **252**) to the user. If the user selects an option that involves data sharing between the host and child application, or with a third party data source, the user may be presented with a notification **254**, as shown in FIG. 2N. FIGS. 2O-2P illustrate a user interface that displays the information shared between host and child applications with the user option to change and/or enter additional data. Examples of such information include applicant name **256**, applicant address **258**, applicant email **260**, applicant phone **262**, applicant social security number **264** and applicant date of birth **266** as shown in FIG. 2O. Examples of such user options include options to choose an applicant user name and password **268**, select statement options **270** and add another authorized user **272**. FIG. 2Q illustrates a user interface that displays terms and conditions **274** as well as a selectable icon **276** for the user to submit approval to the credit provider to use the applicant information. In some embodiments, the child application may concurrently share data with the host application after the user submits approval. FIG. 2R illustrates a user interface indicating that the user has been approved to use credit (e.g., via a Rock Red card). A user electing to use their card immediately may be prompted to enroll their card in a mobile wallet, either within the host application or a third party application (e.g., APPLE PAY®), separate and distinct from the host application or child application, as illustrated in FIG. 2S. FIG. 2T shows the user interface after a successful enrollment in a third party mobile wallet.

(44) In some embodiments, the retailer application **110** may request end-user specific data

associated with the financial institution for a subsequent user interface component to be displayed on a user interface. In these embodiments, the retailer application **110** interfaces with the financial services application **120** through connection **130** to call or request functionality that is not available through the retailer application **110** but, instead, is available from the financial services server **140** of the financial institution (e.g., credit offers) via the financial services application **120**. The retailer application **110** may display that functionality in connection with other functionality of the retailer application **110** (e.g., the cart page or the home page of the retailer application **110**). Thus, embodiments of the present invention allow the user to take advantage of functionality of two separate applications (e.g., the retailer application **110** and those available from the financial services server **140** of the financial institution via the financial services application **120**) in a single user interface, e.g., displayed on mobile device **100**.

(45) For example, from the cart page, retailer application **110** can call a function (e.g., Show Promo function) executed by financial services application **120** to display information from the functionality that is available from the financial services server **140** of the financial institution. Examples of the different call functions are described in more detail below.

(46) FIGS. 3A-3C illustrate exemplary user interfaces for extending functionality of a host application on an electronic device by providing user-specific data (e.g., targeted offers) on the host application using data that is inaccessible to the host application, according to at least some embodiments of the invention.

(47) FIG. 3A is similar to the user interface shown in FIG. 2B, where the user navigates to a user interface of the retailer application **110** showing a shirt that the user would like to purchase. In some embodiments, in response to a user indication that the user desires to buy the item (i.e., the shirt) a retailer may provide an offer or promotion (e.g., \$10 off a purchase) **220** to an end user **150** via the retailer application **110** to use a retailer-specific credit card to complete the transaction. However, the retailer may only desire to provide the offer or promotion under certain conditions such as if certain end-user-specific offer criteria is met. In some embodiments, the end-user-specific offer criteria is based on available credit on a retailer-specific credit card of an end user **150**. For example, as shown in FIG. 3B, the offer or promotion may only be provided if enough is credit available on the retailer-specific credit card account of the end user **150** to complete the transaction. In these embodiments, the retailer application **110** may request the financial services application **120** to retrieve data associated with the retailer-specific credit account (e.g., available credit) of the end user **150**. In response, the financial services application **120** retrieves the available credit information for the end user **150** from financial services server **140**.

(48) In these embodiments, if the retailer application **110** or the financial services application **120** determines that the total amount of items selected by the end user **150** is less than the amount of available credit associated with the retailer-specific credit account of the end user **150** (i.e., that the data associated with the retailer-specific credit account meets end-user-specific offer criteria), then the offer or promotion is displayed to the end user **150** (e.g., on the cart page user interface or front page user interface of the retailer application **110** or a user interface of the financial services application **120**). Given the amount of money to be spent on the items in the cart, the retailer application **110** or financial services application **120** can now return a promotion and ensure that the promotion is valid for someone who has the available credit to spend.

(49) In some embodiments, as shown in the checkout cart user interface illustrated in FIG. 3C, a retailer application **110** may provide an offer or promotion **222** to an end user **150** to increase a credit limit for a retailer-specific credit account to complete a transaction if certain end-user-specific offer criteria is not met. For example, the offer or promotion **222** is only provided if there is not enough credit available on the retailer-specific credit card account of the end user **150** to complete the transaction. In these embodiments, the retailer application **110** may request the financial services application **120** to retrieve data associated with the retailer-specific credit account (e.g., available credit) of the end user **150**. In response, the financial services application **120**

retrieves the available credit information for the end user **150** from financial services server **140**.

(50) In these embodiments, if the retailer application **110** or the financial services application **120** determines that the total amount of items selected by the end user **150** is greater than the amount of available credit associated with the retailer-specific credit card of the end user **150** (i.e., that the data associated with the retailer-specific credit account does not meet the end-user-specific offer criteria), then the offer or promotion to increase the credit limit for the retailer-specific credit card is displayed to the end user **150** (e.g., on the cart page user interface or front page user interface of the retailer application **110** or a user interface of the financial services application **120**). Upon selection to increase the credit limit, the retailer application **110** or the financial services application **120** walks the user through a process to increase the credit limit for the retailer-specific credit card of the end user **150**. This process can be used to ensure that the user has sufficient credit to make the purchase.

(51) While the preceding example uses available credit on a retailer-specific credit account to determine whether certain end-user-specific offer criteria is met, it is contemplated that other criteria may be used to determine whether certain end-user-specific offer criteria is met, including user demographic information and user behavioral information (e.g., purchase history), among others. Other examples of offer criteria include geographic, temporal, climate (i.e., weather), and artificial intelligent software for predicting user behavior.

(52) By leveraging available balance information on a financial account only accessible to the financial services application **120**, the retailer application **110** can provide more-effective targeted offers to the end user **150** without having direct access to personal information (e.g., financial account identification information) of the user. These embodiments improve the security functionality of the computer by protecting the user's personal information from being compromised while also enhancing functionality of the retailer application **110** to provide more-effective targeted offers. These embodiments represent an improvement over conventional applications that merely request the user to provide additional personal details, such as a personal identification information for a financial account, to access additional information (e.g., financial account information) of the user.

(53) Exemplary Process Flow Embodiments

(54) FIG. 4 illustrates a flow diagram depicting a method **400** performed by an electronic device for extending the functionality of a host application based on data that is inaccessible to the host application, in accordance with some embodiments of the invention.

(55) At step **401**, the electronic device (e.g., the client device **100**) invokes a host application executed at the electronic device **100**. In some embodiments, the host application is a retailer application.

(56) At step **402**, the electronic device **100** displays, using the host application **110**, a first host user interface (e.g., the user interface in FIG. 2B, the user interface in FIG. 2G) on the display (e.g., user interface **102**). In some embodiments, the first host user interface includes a first host user interface component (e.g., purchase button **206** in FIG. 2B, credit card link **219** in FIG. 2G).

(57) At step **403**, in response to a selection of the first host UI component in the first host user interface by an end user, the electronic device **100** invokes, using the host application, a child application (e.g., a financial services application **120**) executed at the electronic device **100**.

(58) At step **404**, in further response, the electronic device **100** transmits, using the host application **110**, a request for data associated with a child user interface component to the child application **120**.

(59) At step **405**, in further response, the electronic device **100** transmits, using the child application **120**, the request for data associated with the child user interface component to a child application server **140** connected to the child application **120**. In some embodiments, the data associated with the child user interface component is inaccessible by the host application **110**. In some embodiments, the child application **120** is configured to receive personal information

provided by the end user **150** (e.g., via a previously-displayed child user interface) that is inaccessible by the host application. In some embodiments, the request for data associated with the child user interface component transmitted from the child application includes the personal information provided by the end user. For example, in some embodiments, the personal information is personal identification information or retailer-specific financial account information. (60) At step **406**, in further response, the electronic device **100** receives, using the child application **120**, the data associated with the child user interface component from the child application server **140**. In some embodiments, the data associated with the child user interface component is a child user interface as shown in FIG. 2E or any of FIGS. 2H-2L. In some embodiments, data displayed on the child user interface and data received via the child user interface is inaccessible by the host application. An example of data displayed on the child user interface includes data associated with a financial account of the end user **150**, among others.

(61) At step **407**, in further response, the electronic device **100** displays the child user interface component on the display. In some embodiments, the child user interface component is a child user interface as shown in FIG. 2E and FIGS. 2H-2L. In some embodiments, the child user interface component is a selectable affordance (e.g., a confirm checkout button **218**, a checkout button **220** or a request for a credit increase on a financial account button **222**).

(62) In some embodiments, as shown in exemplary FIGS. 3A-3C, the electronic device **100** provides, using the child application **120**, the data associated with the child user interface component to the host application **110**; and displays, using the host application **110**, the child user interface component on the display. In some embodiments, the host application **110** is a retailer application, the child application **120** is a financial services application and the data associated with a child user interface component includes data associated with a retailer-specific financial account with a credit limit of the end user **150**. In these embodiments, in response to a determination that the data associated with the retailer-specific financial account (e.g., the available credit limit) meets end-user-specific offer criteria, the electronic device **100** displays, using the retailer application **110**, a first offer to the end user to use the retailer-specific credit account to make a purchase (see e.g., FIG. 3B). In some embodiments, in response to a determination that the data associated with the retailer-specific financial account (e.g., the available credit limit) does not meet end-user-specific offer criteria, the electronic device **100** displays, using the retailer application **110**, a second offer to the end user (e.g., to request a credit limit increase for the retailer-specific credit account to make a purchase (see e.g., FIG. 3C)) where the second offer is different from the first offer.

(63) By providing limited access to the data associated with the child user interface component (i.e., financial account information of an end user **150**) using the child application **120**, the end user **150** can avoid exposing additional personal information (e.g., personal identification information for the financial institution) from the host application **110** while still allowing the host application **110** to provide user-specific information (e.g., targeted offers) to the end user **150**. These embodiments represent an improvement to the security capabilities of a computer by limiting exposure of personal information to different parties while still providing enhanced functionality within a computer application.

(64) A computer system that may be used to implement the methods of the present invention, which methods may be implemented as programmable code for execution by computer system, is now described. More particularly, the computer system comprises hardware, as described more fully herein, that is used in connection with executing software/computer programming code (i.e., computer readable instructions) to carry out the steps of the methods described herein.

(65) The computer system includes one or more processors. The processor may be any type of processor, including but not limited to a special purpose or a general-purpose digital signal processor, specially programmed to perform the methods described herein. Processor may be connected to a communication infrastructure (e.g. a data bus or computer network) either via a

wired connection or a wireless connection. Communication infrastructure carries signals and may be implemented using wire or cable, fiber optics, a phone line, a wireless link, a cellular phone link, a radio frequency link, or any other suitable communication channel, including a combination of the foregoing exemplary channels.

(66) The computer system includes one or more memories. The memory may include at least one of: random access memory (RAM), a hard disk drive and a removable storage drive, such as a floppy disk drive, a magnetic tape drive, or an optical disk drive. The removable storage drive reads from and/or writes to a removable storage unit. The removable storage unit can be a floppy disk, a magnetic tape, an optical disk, which is read by and written to a removable storage drive.

(67) In alternative implementations, memory may include other similar means for allowing computer programs or other instructions to be loaded into computer system. Such means may include, for example, a removable storage unit and an interface. Examples of such means may include a removable memory chip (such as an EPROM, or PROM, or flash memory) and associated socket, and other removable storage units and interfaces which allow software and data to be transferred from removable storage unit to the computer system. Alternatively, the program may be executed and/or the data accessed from the removable storage unit, using the processor of the computer system.

(68) The computer system includes one or more user interfaces. The user interface may be a program that controls a display of computer system, on which the output of the processes described herein can be displayed. The user interface may include one or more peripheral user interface components, such as a keyboard or a mouse. The end user may use the peripheral user interface components to interact with computer system. The user interface may receive user inputs, such as mouse inputs or keyboard inputs from the mouse or keyboard user interface components.

(69) Thus, in exemplary embodiments, there is included one or more computers having one or more processors and memory (e.g., one or more nonvolatile storage devices). In some embodiments, memory or computer readable storage medium of memory stores programs, modules and data structures, or a subset thereof for a processor to control and run the various systems and methods disclosed herein. In one embodiment, a non-transitory computer readable storage medium having stored thereon computer-executable instructions which, when executed by a processor, perform one or more of the methods disclosed herein.

(70) It will be appreciated by those skilled in the art that changes could be made to the exemplary embodiments shown and described above without departing from the broad inventive concept thereof. It is understood, therefore, that this invention is not limited to the exemplary embodiments shown and described, but it is intended to cover modifications within the spirit and scope of the present invention as defined by the claims. For example, specific features of the exemplary embodiments may or may not be part of the claimed invention and features of the disclosed embodiments may be combined.

(71) It is to be understood that at least some of the figures and descriptions of the invention have been simplified to focus on elements that are relevant for a clear understanding of the invention, while eliminating, for purposes of clarity, other elements that those of ordinary skill in the art will appreciate may also comprise a portion of the invention. However, because such elements are well known in the art, and because they do not necessarily facilitate a better understanding of the invention, a description of such elements is not provided herein.

(72) Further, to the extent that the method does not rely on the particular order of steps set forth herein, the particular order of the steps should not be construed as limitation on the claims. The claims directed to the method of the present invention should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the steps may be varied and still remain within the spirit and scope of the present invention.

Claims

1. A computer-implemented method, comprising: invoking by one or more processors of an electronic device, a merchant application from a financial services application, wherein the merchant application is associated with a merchant server, and wherein the financial services application is associated with a financial services application server separate from the merchant server; transmitting, by the one or more processors of the electronic device, a request for user data associated with a merchant offer, a purchase, and the financial services application server; encrypting, by the one or more processors, a portion of the user data in accordance with a security model specific to the merchant application; storing, by the one or more processors of the electronic device, the user data in a memory of the electronic device, wherein the user data is stored as secure financial information that is inaccessible to and unmodifiable by the merchant application, wherein the portion of the user data is stored in accordance with a security model of the financial services application server, and wherein the security model excludes access by the merchant application to the secure financial information; facilitating, by the one or more processors of the electronic device, communication between the financial services application and the merchant application; and receiving, by the one or more processors of the electronic device, an offer based on the secure financial information and purchase data for the purchase, wherein when the offer is received, the electronic device maintains the secure financial information as inaccessible and unmodifiable by the merchant application.
2. The computer-implemented method of claim 1, further comprising: presenting a shared user interface on a display of the electronic device, wherein the shared user interface includes a user option to change or enter additional data, and wherein control of the shared user interface is transferrable between the merchant application and the financial services application.
3. The computer-implemented method of claim 1, further comprising: receiving, at the electronic device, a request to execute the merchant application; and receiving, at the electronic device and via the merchant application, user input data authorizing communications to facilitate the purchase including secure communications with the financial services application server and the merchant server.
4. The computer-implemented method of claim 1, wherein the financial services application is encoded so that the electronic device is prevented from modifying a code of the financial services application, and wherein the secure financial information is maintained as inaccessible and unmodifiable by the electronic device by encrypting the secure financial information using the financial services application in accordance with the security model specific to the financial services application and the financial services application server.
5. The computer-implemented method of claim 1, further comprising: generating a device identifier for authenticating a user associated with the electronic device; and storing the device identifier in an encrypted storage of the electronic device, wherein the financial services application is invoked using device identifier data matching the device identifier stored in the encrypted storage of the electronic device.
6. The computer-implemented method of claim 1, wherein the offer includes a credit increase offer, wherein the secure financial information includes financial account details, and wherein the purchase data includes a purchase price.
7. The computer-implemented method of claim 1, wherein the offer includes a discount offer, wherein the secure financial information includes an available credit amount, and wherein the discount offer corresponds to a purchase range matching the available credit amount.
8. An electronic device, comprising: a memory; and one or more processors coupled to the memory and configured to perform operations including: invoking by the one or more processors of the electronic device, a merchant application from a financial services application, wherein the

merchant application is associated with a merchant server, and wherein the financial services application is associated with a financial services application server separate from the merchant server; transmitting, by the one or more processors of the electronic device, a request for user data associated with a merchant offer, a purchase, and the financial services application server; encrypting, by the one or more processors, a portion of the user data in accordance with a security model specific to the merchant application; storing, by the one or more processors of the electronic device and in the memory, the user data as secure financial information that is inaccessible to and unmodifiable by the merchant application, wherein the portion of the user data is stored in accordance with a security model of the financial services application server, and wherein the security model excludes access by the merchant application to the secure financial information; facilitating, by the one or more processors of the electronic device, communication between the financial services application and the merchant application; and receiving, by the one or more processors of the electronic device, an offer based on the secure financial information and purchase data for the purchase, wherein when the offer is received, the electronic device maintains the secure financial information as inaccessible and unmodifiable by the merchant application.

9. The electronic device of claim 8, wherein the one or more processors are configured to perform operations further comprising: presenting a shared user interface on a display of the electronic device, wherein the shared user interface includes a user option to change or enter additional data.

10. The electronic device of claim 8, wherein the one or more processors are configured to perform operations further comprising: receiving, at the electronic device, a request to execute the merchant application; and receiving, at the electronic device via the merchant application, user input data authorizing communications to facilitate the purchase including secure communications with the financial services application server and the merchant server.

11. The electronic device of claim 8, wherein the financial services application is encoded so that the electronic device is prevented from modifying a code of the financial services application, and wherein the secure financial information is maintained as inaccessible and unmodifiable by the electronic device by encrypting the secure financial information using the financial services application in accordance with the security model specific to the financial services application and the financial services application server.

12. The electronic device of claim 8, wherein the one or more processors are configured to perform operations further comprising: generating a device identifier for authenticating a user associated with the electronic device; and storing the device identifier in an encrypted storage of the electronic device, wherein the financial services application is invoked using device identifier data matching the device identifier stored in the encrypted storage of the electronic device.

13. The electronic device of claim 8, wherein the offer includes a credit increase offer, wherein the secure financial information includes financial account details, and wherein the purchase data includes a purchase price.

14. The electronic device of claim 8, wherein the offer includes a discount offer, wherein the secure financial information includes an available credit amount, and wherein the discount offer corresponds to a purchase range matching the available credit amount.

15. A non-transitory computer readable medium comprising instructions that, when executed by one or more processors of an electronic device, cause the electronic device to perform operations comprising: invoking by the one or more processors of the electronic device, a merchant application from a financial services application, wherein the merchant application is associated with a merchant server, and wherein the financial services application is associated with a financial services application server separate from the merchant server; transmitting, by the one or more processors of the electronic device, a request for user data associated with a merchant offer, a purchase, and the financial services application server; encrypting, by the one or more processors, a portion of the user data in accordance with a security model specific to the merchant application; storing, by the one or more processors of the electronic device, the user data in a memory of the

electronic device, wherein the user data is stored as secure financial information that is inaccessible to and unmodifiable by the merchant application, wherein the portion of the user data is stored in accordance with a security model of the financial services application server, and wherein the security model excludes access by the merchant application to the secure financial information; facilitating, by the one or more processors of the electronic device, communication between the financial services application and the merchant application; and receiving, by the one or more processors of the electronic device, an offer based on the secure financial information and purchase data for the purchase, wherein when the offer is received, the electronic device maintains the secure financial information as inaccessible and unmodifiable by the merchant application.

16. The non-transitory computer readable medium of claim 15, wherein when executed by the one or more processors, the instructions cause the one or more processors to perform operations further comprising: presenting a shared user interface on a display of the electronic device, wherein the shared user interface includes a user option to change or enter additional data.

17. The non-transitory computer readable medium of claim 15, wherein when executed by the one or more processors, the instructions cause the one or more processors to perform operations further comprising: receiving, at the electronic device, a request to execute the merchant application; and receiving, at the electronic device via the merchant application, user input data authorizing communications to facilitate the purchase including secure communications with the financial services application server and the merchant server.

18. The non-transitory computer readable medium of claim 15, wherein the financial services application is encoded so that the electronic device is prevented from modifying a code of the financial services application, and wherein the secure financial information is maintained as inaccessible and unmodifiable by the electronic device by encrypting the secure financial information using the financial services application in accordance with the security model specific to the financial services application and the financial services application server.

19. The non-transitory computer readable medium of claim 15, wherein when executed by the one or more processors, the instructions cause the one or more processors to perform operations further comprising: generating a device identifier for authenticating a user associated with the electronic device; and storing the device identifier in an encrypted storage of the electronic device, wherein the financial services application is invoked using device identifier data matching the device identifier stored in the encrypted storage of the electronic device.

20. The non-transitory computer readable medium of claim 15, wherein the offer includes a credit increase offer, wherein the secure financial information includes financial account details, and wherein the purchase data includes a purchase price.

21. The non-transitory computer readable medium of claim 15, wherein the offer includes a discount offer, wherein the secure financial information includes an available credit amount, and wherein the discount offer corresponds to a purchase range matching the available credit amount.
